

AI Prompt Documentation

KOHLER AI Bathroom Designer & Planner — Track 1 Case Study

This document captures the AI prompts that shaped the design and functionality of the app — the recommendation engine, product catalog logic, interactive viewport, and comparison engine — along with what each prompt produced. Routine one-off bug fixes (e.g. a button losing its color in dark mode, a popup not appearing, minor spacing tweaks) and repository/setup logistics have been left out to keep this focused on the substantive product and design decisions.

1. Budget + Size Driven Recommendation Logic

PROMPT

Is it possible to add a variable like — I ask for a budget, for example \$4000, and 6 by 8 feet with smart privacy glass — and it gives me all the possible combinations and recommendations?

OUTCOME

Built a lightweight rule-based parser (regex-driven, no external AI API required) that scans free-text input for budget figures, room dimensions, and add-on keywords (privacy glass, mirror, lighting, shower), then computes a matching recommendation on the fly. This became the core `generateAIResponse()` function that powers the AI chat panel. The parser reads the user's sentence the way a search engine scans for patterns, using regular expressions for currency figures (e.g. "\$4000"), dimension pairs (e.g. "6 by 8"), and keyword flags for each add-on, rather than matching one fixed sentence structure.

2. Generalizing the Parser Beyond One Example

PROMPT

But can't I add other values — like \$8000 and 8 by 12 with ambient lighting — and have it still work?

OUTCOME

Confirmed and demonstrated that because the logic uses regex pattern matching rather than hardcoded phrases, it generalizes to any budget, dimension, and add-on combination the user types — not just the original example. Walked through the exact parsing steps for a new input ("8000 dollars and 8 by 12 with ambient lighting"): detect the number, detect the dimension pair, detect the lighting keyword, subtract the add-on cost from the budget, and check the remainder against the tier thresholds.

3. Full Recommendation Output (Tier + Theme + Total Cost)

PROMPT

I want all the recommendations — I specify my budget, my size and my AI add-ons, and it does the calculations and recommends not only the tier but also which architectural type, as well as the total cost including add-ons and everything.

OUTCOME

Extended the recommendation engine to calculate a dynamic size multiplier from the entered square footage, tally selected add-ons, pick a matching tier and architectural theme, and print a full 'AI Blueprint' receipt with a grand total — turning a simple budget check into a complete, itemized recommendation.

4. Multiple Recommendations, Not Just One

PROMPT

Does this code give only one recommendation or multiple based on the parameters provided?

OUTCOME

Identified that the engine was only returning the single highest tier that fit the budget. Updated the logic to evaluate all three tiers (Value, Balanced, Premium) and return every option that qualifies under the given budget, giving the user real choices to compare instead of one fixed answer.

5. Style-Aware Filtering and Budget-Exceeded Handling

PROMPT

Update the code for all the add-ons — also, when the budget limit is exceeded, show no available options. On top of that, use words like: an imperial feel = Japanese Zen, minimalist is Modern, and luxurious or old money = Classic Luxury. Add keywords that indicate the specific style if mentioned; if not indicated, show all the recommendations.

OUTCOME

Added keyword-based style detection so the AI chat could recognize informal descriptions ("old money", "imperial", "minimalist") and map them to the app's actual architectural themes (Classic Luxury, Japanese Zen, Modern). Integrated all four smart add-ons into the cost math, filtered recommendations by the detected style when one was mentioned, and added a strict "No available options" message when the budget couldn't cover any tier.

6. Per-Product Detail View

PROMPT

Is it possible to add detailed information about each individual vanity, shower, toilet, and faucet — like you click it and then you can see more detail on each product?

OUTCOME

Turned static product cards into clickable elements that open a frosted-glass modal (lightbox) showing that product's finish, eco-status, and description, without navigating away from the main planner view. Cards also elevate slightly on hover to signal they're interactive.

7. Interactive Floor Plan Viewport

PROMPT

Is it possible to make the viewport interactive, in the sense where the bathroom image that's generated can be zoomed in and out and rotated on that plane, clockwise and anticlockwise?

OUTCOME

Replaced the static schematic canvas with a fully interactive viewport: a frosted-glass control overlay for zoom and 90-degree rotation (left/right and reset), mouse-wheel zoom, and click-and-drag panning across the generated floor plan.

8. Repositioning the Cost Summary for a Premium Layout

PROMPT

Instead of the total appearing in the side menu, I want it to just have the AI selections, and at the bottom below the viewport it will give the total configuration summary with add-on prices.

OUTCOME

Moved the cost breakdown out of the cramped sidebar and into a full-width itemized summary panel below the 3D/schematic viewport, styled like a premium receipt — separating product selection (sidebar) from cost review (bottom panel) for a cleaner, more professional configurator layout.

9. Side-by-Side Comparison Engine

PROMPT

For the last major feature, I want a comparison menu — click the compare icon to reach a page with two configs, left and right, to compare. Choose model, dimensions and architecture type on each side, and generate a summary at the bottom comparing the two (showing both viewports) with the advantages of each.

OUTCOME

Built a full-screen 'Comparison Mode' overlay with independently configurable Room Size, Theme, and Tier selections on each side, rendering two floor-plan viewports side by side, plus an algorithm that auto-generates the objective advantages of each configuration — the app's single most complex feature.

10. Polishing the Comparison Experience

PROMPT

First UI change: when entering the comparison panel, I want the Kohler zoom-in loading transition to happen before we enter. Second: the comparison panels of bathrooms should be more glass-like. Third: on the main menu there are too many colors on the icons for volume, assistant, compare, and dark/light mode — make the icons monochrome.

OUTCOME

Applied the branded loading transition to the entry into comparison mode for a consistent, cinematic feel; restyled the comparison panels with a frosted-glass aesthetic to match the rest of the app; and replaced the colorful emoji icons in the utility bar with a clean, monochrome icon set for visual consistency.

All development was carried out through iterative conversational prompting with an AI assistant (Gemini), with the AI's code proposals reviewed, tested locally, and refined across each iteration.